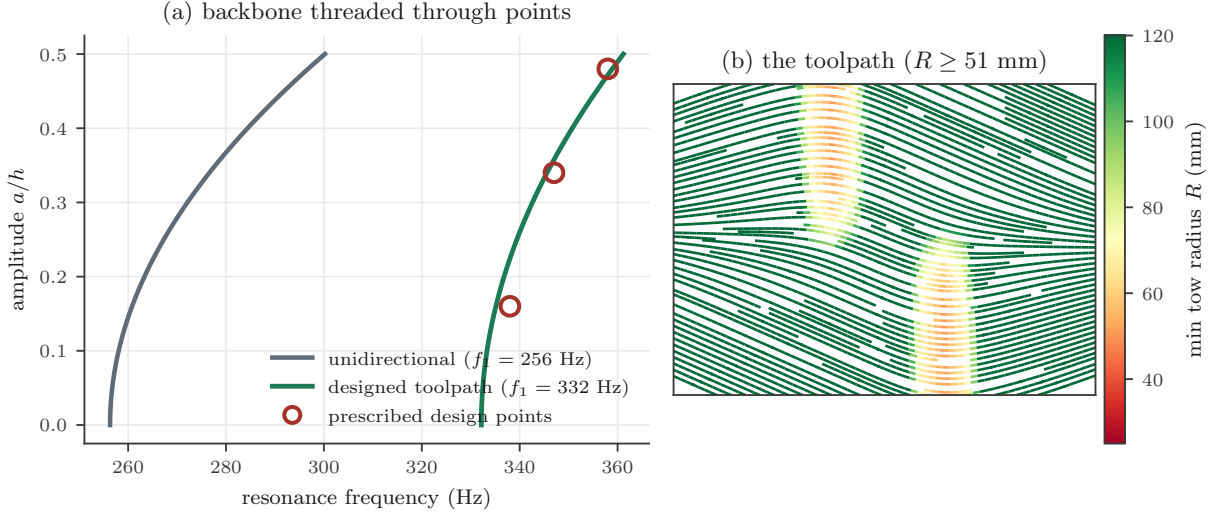

(a) backbone threaded through points
amplitude a/h
0.0
0.1
0.2
0.3
0.4
0.5
260
280
300
320
340
360
resonance frequency (Hz)
unidirectional ($f_1 = 256$ Hz)
designed toolpath ($f_1 = 332$ Hz)
prescribed design points
(b) the toolpath ($R \geq 51$ mm)
120
100
80
60
40
min tow radius R (mm)